# Kelvin Xu

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## Education

**Northeastern University** — Boston, MA
*Bachelor of Science in Computer Science, GPA: 3.86/4.00* — *September 2021 – May 2025*

## Experience

**Software Engineer Co-op** — January 2024 – June 2024
*Transperfect Legal Solutions* — *Maynard, MA*

- Collaborated within a small development team to build and enhance an industry-leading e-discovery and early case assessment management platform, leveraging Java, PostgreSQL, and Hibernate for backend development.
- Redesigned the import/export process for stored client project data in the application, improving compatibility across devices and enabling seamless migration to newer versions while increasing efficiency and speed.
- Initiated and prototyped integration of the DocuSign API, enabling clients to manage and sign documents directly within the application without relying on external software.

**Quality Assurance Engineer Co-op** — January 2023 – August 2023
*Transperfect Legal Solutions* — *Maynard, MA*

- Developed automated front-end test suites in Java using Selenium and JUnit to detect bugs and validate performance across multiple interfaces, applying object-oriented design principles to ensure modular code.
- Resolved 15+ development tickets, collaborating closely with team members through peer reviews and clear communication, and producing thorough documentation to support implementation and future maintenance.

## Projects

**OpenGL Rendering Engine** | *C++, OpenGL, CMake, Git* — March 2025 – April 2025

- Created a real-time graphics engine supporting both rasterization and ray tracing using OpenGL and C++.
- Implemented a modular scenegraph architecture to manage hierarchical transformations, define light sources, and enable keyframe-based animations with interactive scene rotation for enhanced visualization and control.
- Added support for texture mapping and bump mapping for enhanced surface detail and graphical fidelity.

**Spreadsheet Application** | *TypeScript, React, Express.js, Jest, Git* — November 2024 – December 2024

- Developed a TypeScript-based spreadsheet application with a React front-end, enabling core functionalities like cell editing, complex formula calculations, cell referencing, row and column resizing, and error handling.
- Implemented advanced features, including customizable graph creation for various graph types, a custom formula builder, cell history, and version history, enhancing user flexibility and workflow.
- Followed Test-Driven Development and Object-Oriented Design principles, utilizing design patterns such as the visitor pattern, singletons, and generics for maintainability and scalability.

**QuickEats** | *Python, Flask, MySQL, Ngrok, AppSmith* — October 2024 – November 2024

- Designed and implemented an online food delivery service in Python, modeling system interactions with UML to support order management and driver dispatch.
- Extended the system with a front-end AppSmith application, deployed using Docker containers with a Flask REST API and MySQL database, with Ngrok enabling seamless client-to-API communication.

**Distritubed Key-Value Database** | *Python* — December 2023 – December 2023

- Engineered a distributed, replicated key-value database using Python.
- Integrated the Raft consensus protocol to ensure strong consistency across nodes, even under conditions of network unreliability, packet loss, and server failures.
- Optimized the system for low-latency client responses and minimal packet transmission, achieving high efficiency in distributed coordination and fault tolerance.

## Technical Skills

**Languages**: Java, Python, C#, C/C++, TypeScript, JavaScript, SQL, HTML/CSS, R, Assembly (x86-64)
**Frameworks**: React, Node.js, Express.js, Flask, Spring Boot, Hibernate
**Developer Tools**: Git, Docker, IntelliJ, Unity, VS Code, Visual Studio, CMake, BitBucket, JIRA, Confluence
**Libraries**: OpenGL, JUnit, Mockito, Jest, NumPy, Pandas, Newtonsoft.Json, Nearley